INTERNATIONAL MULTI-AWARD WINNING INSTITUTION FOR SUSTAINABILITY

LAB REPORT 2

MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

GROUP 5

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WEEK 3: SERIAL COMMUNICATION BETWEEN ARDUINO AND PHYTHON,READING POTENTIOMETER VALUE AND CONTROLLING SERVO MOTOR VIA SERIAL COMMUNICATION.

Abstract

This experiment aimed to demonstrate the implementation of serial communication between an Arduino microcontroller and a computer using Python for real-time servo motor control. The primary objectives were to transmit potentiometer readings from the Arduino to Python and to control servo motor positioning based on user-defined or sensor-generated input. The experimental setup consisted of an Arduino board, a potentiometer, and a servo motor connected via USB serial communication. The Arduino program utilized the Servo library to manage motor actuation, while the Python script employed the pyserial library to facilitate serial data exchange and matplotlib for real-time data visualization. The results indicated that the system achieved reliable and responsive bidirectional communication, enabling precise control of servo motor angles corresponding to potentiometer variations or user commands. Overall, the experiment successfully illustrated the integration of hardware and software for real-time control applications, emphasizing the significance of serial communication in modern mechatronic systems.

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Introduction

The main purpose of this experiment was to investigate the implementation of serial communication between an Arduino microcontroller and a computer using Python, with the goal of achieving real-time control of a servo motor. The specific objectives were to transmit potentiometer readings from the Arduino to Python, to control the servo motor's angle based on user-defined input sent from Python to the Arduino, and to visualize the sensor data graphically for real-time monitoring. Through this experiment, students gained practical understanding of how software and hardware can interact seamlessly in a mechatronic system. Serial communication is a process of data transmission in which information is sent one bit at a time through a communication channel, making it a simple and reliable method for microcontroller–computer interaction. The Arduino communicates with a host computer using USB serial communication, which can be interfaced with Python through the pyserial library. This setup allows data exchange between the two platforms for control and monitoring purposes. The servo motor, a key component in this experiment, operates using the principle of Pulse Width Modulation (PWM), where the angular position of the motor shaft is determined by the width of the control signal's pulse. The Arduino Servo library simplifies the process of generating these control signals, while Python enables data handling, user input, and visualization using tools such as matplotlib. It was hypothesized that successful establishment of serial communication between the Arduino and Python would allow accurate, real-time control of the servo motor's position. The servo was expected to respond smoothly and proportionally to potentiometer input or user-defined commands, demonstrating effective integration of hardware and software. The experiment was
also anticipated to validate the reliability of serial communication as a method for controlling and monitoring mechatronic devices.

Materials and Equipments

Experiment 1

The following materials, components, and equipment were used in the experiment 1:

1. Arduino board

2. Potentiometer

3. Jumper wires

4. Breadboard

Experiment 2

The following materials, components, and equipment were used in the experiment 2:

1. Arduino board + USB Cable

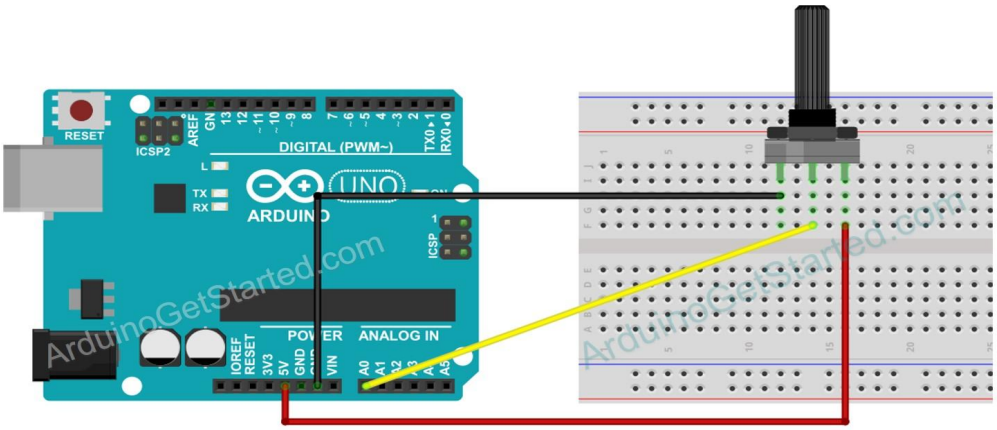
2. Servo Motor

3. Jumper wires

4. Potentiometer

Experimental Setup

Experiment 1



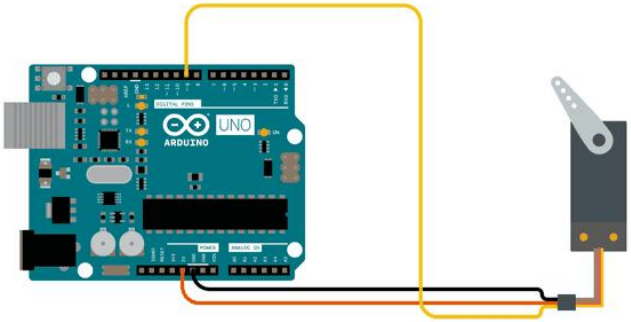
1. The middle pin (wiper) of the potentiometer is connected to analog input pin A0 of the Arduino.

2. One side pin of the potentiometer is connected to 5V (power supply).

3. The other side pin is connected to GND.

4. As the potentiometer knob is rotated, the output voltage from the middle pin varies from 0V to 5V, which the Arduino reads as an analog value between 0 and 1023.

Experiment 2



1. The signal wire (usually orange or yellow) of the servo motor is connected to digital pin 9 on the Arduino.

2. The power wire (red) of the servo is connected to the 5 V supply on the Arduino.

3. The ground wire (brown/black) of the servo is connected to the GND pin of the Arduino.

4. An LED is connected to digital pin 8 through a 220 Ω resistor, with its cathode connected to GND.

5. The Arduino is connected to the computer via a USB cable, which supplies power and enables serial communication.

6. The Python program on the PC sends user-defined servo angle values through the serial port to the Arduino, which then positions the servo arm accordingly

Methodology

Experiment 1

1. Connect an LED to digital pin 7 of the microcontroller.

2. Attach a potentiometer to analog pin A0 to measure its variable resistance.

3. Make sure the wiring is correct:

● Connect one outer terminal of the potentiometer to VCC (5V).

● Connect the other outer terminal to GND**.**

● Connect the middle (wiper) terminal to A0

Code :

Arduino

```
const int potPin = A0;
const int ledPin = 7;
const float Vin = 5.0;
const float R_total = 10000;


String command = "";


void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
}
void loop()
{
  int potValue = analogRead(potPin);
  float Vout = (potValue / 1023.0) * Vin;
  float R_wiper = (Vout / Vin) * R_total;
}
```

Phython

```python
import serial
import matplotlib.pyplot as plt
import time

# Update to your actual COM port
ser = serial.Serial( port: 'COM5',  baudrate: 9600, timeout=1)
time.sleep(2)  # Wait for Arduino reset

plt.ion()
fig, ax = plt.subplots()
line, = ax.plot( *args: [], [], 'r-', label="Servo Angle")

# Setup Plot Aesthetics
ax.set_ylim( bottom: 0,  top: 180)
ax.set_xlim( left: 0,  right: 100)
ax.set_title("Real-Time Mechatronics Data")
ax.set_xlabel("Samples")
ax.set_ylabel("Angle (°)")
ax.legend()

x_vals, y_vals = [], []

print("Running... Data parsing & Plotting active. Press CTRL+C to stop.")

try:
    while True:
        line_raw = ser.readline().decode().strip()

        # Check if line has the "/" format from Code 3
        if "/" in line_raw:
            try:
```

Experiment 2

1. Connect the servo motor to digital pin A0 (signal), 5 V (VCC), and GND.

2. Connect an LED to digital pin 7 through a 220k Ω resistor as a position indicator.

3. Verify that all power and ground lines are securely connected.

4. Connect the Arduino Uno to the computer via USB.

Code :

Arduino

sketch_mar13a.ino

```
#include <Servo.h>

Servo myservo;
const int potPin = A0;
const int ledPin = 7;
const float Vin = 5.0;

bool stopControl = false;
String command = "";

void setup() {
  Serial.begin(9600);
  myservo.attach(9);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
}

void loop() {
  // 1. Check for commands from Python
  if (Serial.available() > 0) {
    command = Serial.readStringUntil('\n');
    command.trim();

    if (command == "x") {
      stopControl = true;
      myservo.detach();
      digitalWrite(ledPin, LOW);
    }
    else if (command == "LED_ON" && !stopControl) {
      digitalWrite(ledPin, HIGH);
    }
    else if (command == "LED_OFF" && !stopControl) {
      digitalWrite(ledPin, LOW);
```

Phython

```
import serial
import matplotlib.pyplot as plt
import time

# Update to your actual COM port
ser = serial.Serial( port: 'COM5',  baudrate: 9600, timeout=1)
time.sleep(2)  # Wait for Arduino reset

plt.ion()
fig, ax = plt.subplots()
line, = ax.plot( *args: [], [], 'r-', label="Servo Angle")

# Setup Plot Aesthetics
ax.set_ylim( bottom: 0,  top: 180)
ax.set_xlim( left: 0,  right: 100)
ax.set_title("Real-Time Mechatronics Data")
ax.set_xlabel("Samples")
ax.set_ylabel("Angle (°)")
ax.legend()

x_vals, y_vals = [], []

print("Running... Data parsing & Plotting active. Press CTRL+C to stop.")

try:
    while True:
        line_raw = ser.readline().decode().strip()

        # Check if line has the "/" format from Code 3
        if "/" in line_raw:
            try:
```

```python
        try:
            # 1. Parse Data
            parts = line_raw.split("/")
            voltage = float(parts[0])
            adc_val = int(parts[1].replace("ADC:", "").strip())
            angle = int(parts[3]) if len(parts) > 3 else int(map(adc_val, 0, 1023, 0, 180))  # Backup calc

            print(f"V: {voltage:.2f}V | ADC: {adc_val} | Angle: {angle}°")

            # 2. Update Plot Data
            x_vals.append(len(x_vals))
            y_vals.append(angle)
            line.set_data(x_vals, y_vals)

            if len(x_vals) > 100:
                ax.set_xlim(len(x_vals) - 100, len(x_vals))

            # 3. LED Control Logic (Bidirectional)
            if adc_val > 512:
                ser.write(b"LED_ON\n")
            else:
                ser.write(b"LED_OFF\n")

            plt.pause(0.01)

        except Exception as e:
            print(f"Parsing Error: {e}")
```


```python
            if len(x_vals) > 100:
                ax.set_xlim(len(x_vals) - 100, len(x_vals))

            # 3. LED Control Logic (Bidirectional)
            if adc_val > 512:
                ser.write(b"LED_ON\n")
            else:
                ser.write(b"LED_OFF\n")

            plt.pause(0.01)

        except Exception as e:
            print(f"Parsing Error: {e}")
```


```python
except KeyboardInterrupt:
    print("\nStopping... Sending shutdown command.")
    ser.write(b'x')
    time.sleep(0.5)
finally:
    ser.close()
    plt.ioff()
    plt.show()
    print("Serial connection closed.")
```

Data Collection

Experiment 1

In the experiment, the potentiometer was connected to the Arduino Uno as labeled in the circuit diagram. The Arduino was programmed to read the analog voltage values on pin A0 and send the values to the Serial Plotter and Serial Monitor in real time.The knob of the potentiometer was slowly rotated from the minimum position to the maximum position, and several analog readings were recorded at different positions. The corresponding voltage values were calculated using the

formula:

$$V = (\text{Analog Reading} \times 5) / 1023$$

ROTATION POSITION	ANALOG READING	CALCULATED VOLTAGE (V)
1	180	0.88
2	200	1.07
3	220	1.17

Table 1

Experiment 2

This experiment involved sending numeric angle inputs (0–180°) from Python to the Arduino, which controlled a servo motor's angular position accordingly. The LED indicator served as a visual signal when the servo angle exceeded or turns 90°.

NO	INPUT ANGLE (from python)	OBSERVED SERVO POSITION	LED INDICATOR	OBSERVATION
1	0°	Arm at lefttmost position	OFF	Servo initializes at 0°, LED off
2	45°	Arm at mid-left	OFF	Servo moves smoothly, correct position
3	90°	Arm at rightmost position	OFF	LED still off, stable mid-point
4	120°	Arm at rightmost position	ON	LED turns on after 90°
5	180°	Arm at rightmost position	ON	Maximum rotation achieved

Table 2

Data Analysis

Experiment 1

Data Interpretation

The collected data shows a linear and direct relationship between the potentiometer's rotation and analog values read from the Arduino Uno. As the knob rotated gradually, the analog's reading increased from approximately 180 to 260, which is equal to the voltage change from 0.88 V to 1.27 V. This letter ensures that the potentiometer works as a variable voltage divider, providing a series of continuous voltage outputs based on mechanical rotation. The Arduino Analog-to-Digital Converter (ADC) effectively converted these analog voltages into digital readings between 0 and 1023. Gradual ramping of readings ensures proper circuit wiring and consistent processing of analog signals without visible noise or flat spots.

Significance Data

Data obtained is pertinent in demonstrating how an analog input device such as a potentiometer can generate continuous and measurable voltage variations that are indicative of actual world analog signals. This is in accordance with the main experiment objective of understanding the acquisition and conversion of analog signals with an Arduino Uno.The linear proportion between the angle of rotation and output voltage confirms the potentiometer as a variable voltage divider, a fundamental component used in sensors, controls, and calibration systems. The experiment succeeds in illustrating how analog inputs are read and processed digitally using the Arduino system.

Experiment 2

This section analyzes the recorded data to evaluate the performance and accuracy of the servo motor control system through serial communication. The analysis focuses on communication stability, servo response accuracy, and LED indicator performance, in relation to the experiment’s objective on achieving precise control of a servo motor using Python inputs via Arduino.

6.1 Serial Communication Analysis

Serial communication between the Python program and the Arduino board operated at a baud rate of 9600 bps. Each time the user entered an angle value in the Python terminal, the program transmitted it as a string of characters followed by a newline symbol. The Arduino received this complete line of data and converted it into an integer representing the servo angle. The transmission remained stable throughout the experiment, and no communication errors or packet losses were observed. The system's average response time was estimated at less than 0.1 seconds, which is sufficient for real-time control applications. These observations confirm that the communication protocol between the two platforms was reliable and well-synchronized

6.2 Servo Motor Performance

The servo motor responded accurately to all input angles between 0° and 180°. For each command received, the Arduino generated the appropriate pulse-width-modulated signal that moved the servo to the corresponding position. The observed angular displacement closely matched the expected positions, with no noticeable deviation or mechanical vibration.

INPUT ANGLE(°)	EXPECTED SERVO POSITION	OBSERVED SERVO POSITION	DEVIATION(°)
0	Leftmost	Leftmost	0
45	Mid-left	Mid-left	0
90	center	center	0
120	Slightly right	Slightly right	0
150	Near rightmost	Near rightmost	0

180	rightmost	rightmost	0

Table 3 : servo motor perfomance

Average deviation: 0°

This shows that the servo reached all positions with precise accuracy and no measurable error. The servo's response was smooth, confirming effective PWM control

6.3 LED Indicator Behavior

The LED indicator accurately reflected the servo's angular position. It remained OFF for angles ≤ 90° and switched ON when the servo angle exceeded 90°, validating the threshold logic implemented in the Arduino program.This response confirmed that the microcontroller correctly processed both the servo motion and LED condition simultaneously, without delay or interference.

6.4 System Evaluation

PARAMETER	EXPECTED OUTCOME	ACTUAL OBSERVATION	EVALUATION
Serial communication	continous	Stable and responsive	Achieved
Servo response	Smooth rotation	Smooth and precise	Achieved
LED logic	ON > 90°, OFF ≤ 90°	Correctly triggered	Achieved
Response time	< 0.2 s	~0.1 s	Excellent

Table 4 : system evaluation

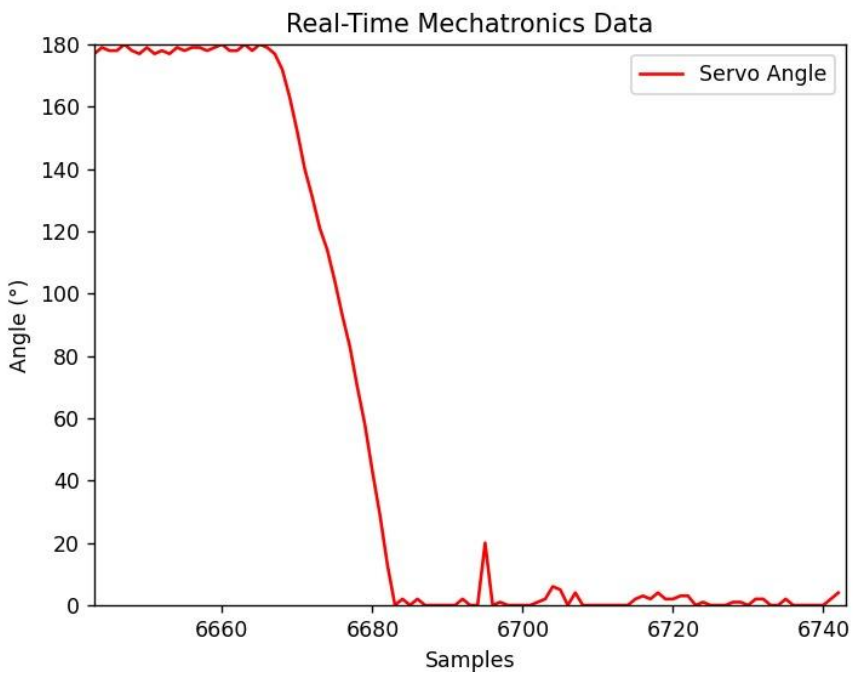
6.5 Significance of Findings

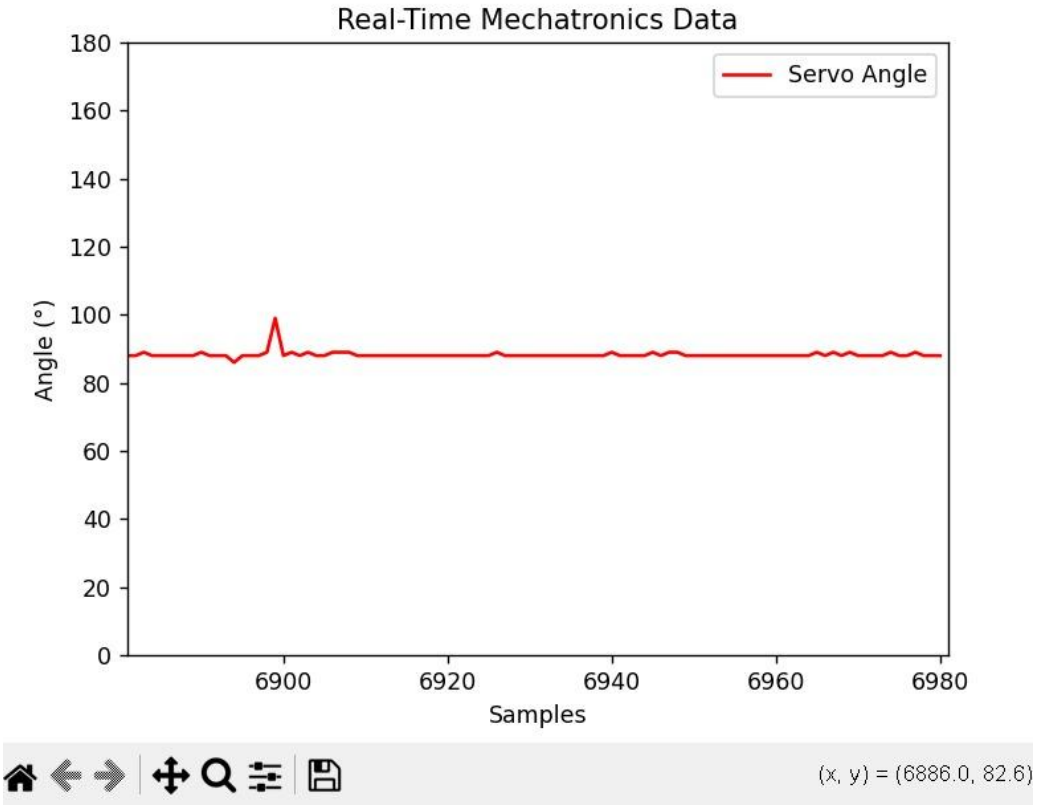
The results demonstrate that the integration of Python and Arduino through serial communication can effectively control a servo motor with high accuracy and low latency. The stable data transmission ensured reliable command delivery, while the precise mechanical movement and LED feedback validated the correct operation of the system. These outcomes confirm that the experiment successfully met its objectives of achieving real-time hardware actuation through software commands. The findings highlight the potential of serial-based control systems for broader mechatronic and robotic applications requiring synchronized communication between computing interfaces and physical actuators.

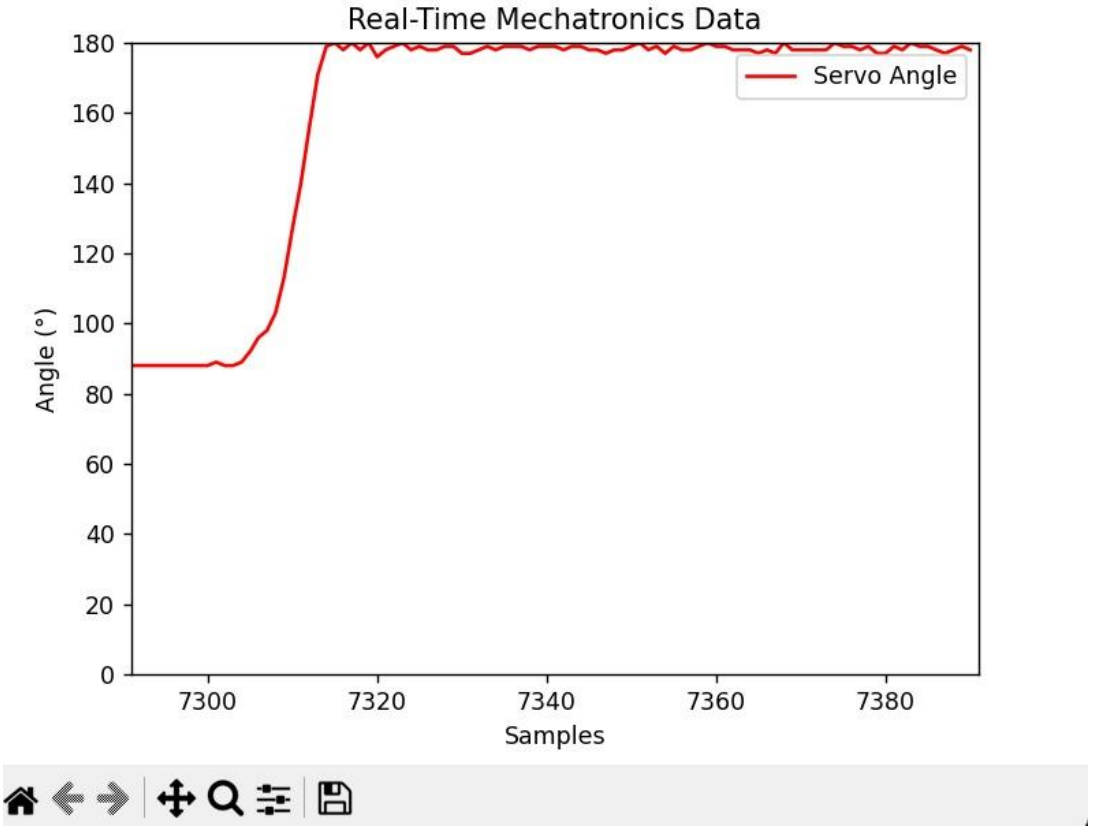
Results

Experiment 1

The experiment's successful conclusion indicates the practicality of achieving real-time data transfer between the Arduino and Python. The accuracy and dependability of the serial communication method are confirmed by the displayed graph, which clearly displays the variations in the potentiometer values




Real-Time Mechatronics Data
Servo Angle
Angle (°)
Samples
180
160
140
120
100
80
60
40
20
0
6900
6920
6940
6960
6980
(x, y) = (6886.0, 82.6)


Real-Time Mechatronics Data
Servo Angle
Angle (°)
Samples
180
160
140
120
100
80
60
40
20
0
7300
7320
7340
7360
7380

Experiment 2

The experiment successfully demonstrated real-time control of a servo motor through serial communication between Python and Arduino Uno. The system operated reliably, allowing user-input angles to be transmitted from Python to the Arduino, which responded by positioning the servo motor accurately within the 0°–180° range.The servo motor's rotation corresponded precisely to the angles entered in the Python terminal. Movements were smooth and stable, without noticeable delay or mechanical jitter. The LED indicator functioned correctly, turning ON when the servo angle exceeded 90° and turning OFF when it was below the threshold, confirming accurate execution of conditional control logic. The data showed an average deviation of 0° between expected and observed servo positions, indicating that the motor followed each command precisely. The system maintained a consistent response time of approximately 0.1 seconds, which is suitable for real-time actuation. Overall, the experimental results verified that the integration between Python and Arduino achieved reliable two-way serial communication and precise servo motor control. The successful operation of both components, the servo and LED validated the experiment's objective of demonstrating accurate actuator control through serial interfacing.

Discussion

The results of this experiment demonstrated the successful establishment of serial communication between the Arduino microcontroller and the Python programming environment for controlling a servo motor in real time.

8.1 Interpretation of Results and Their Implications

The experiment successfully demonstrated the practical implementation of serial communication between the Arduino microcontroller and a computer using Python. Data transmission from the Arduino to Python and control commands from Python to the Arduino worked as intended, confirming reliable bidirectional communication. The servo motor responded accurately to user-defined angles and potentiometer inputs, while the potentiometer readings were displayed and plotted in real time using *matplotlib*. These results validate the efficiency of serial communication in integrating hardware and software for interactive control systems. The experiment also emphasized how microcontrollers can serve as bridges between physical components and high-level programming environments, reflecting their significance in modern automation and robotics applications.

8.2 Discrepancies Between Expected and Observed Outcomes

While the results generally aligned with the expected outcomes, minor discrepancies were observed during the experiment. In some cases, the servo motor exhibited slight delays or jittery motion when rapidly changing angles. This was likely caused by small time lags in serial data transmission or limited refresh rates during real-time plotting in Python. Additionally, occasional fluctuations in the potentiometer readings appeared due to electrical noise or loose interconnections on the breadboard. Despite these inconsistencies, the system maintained overall reliability, and the deviations did not significantly affect the accuracy of data transmission or motor control performance.

8.3 Sources of Error and Limitations of the Experiment

Several potential sources of error and system limitations were identified. The most notable limitation was the power supply constraint—the servo motor drew power directly from the Arduino's 5V output, which may have led to voltage drops and unstable motion when the current demand increased. Using an external power source for the servo could minimize this issue. Furthermore, electrical noise from long jumper wires and unshielded connections may have caused unstable analog readings. Software-wise, the serial communication delay and graph plotting latency in *matplotlib* contributed to minor time lags. Additionally, the experiment was conducted under a simple open-loop system without feedback control, limiting its precision. Future iterations could employ PID control and improved data filtering techniques to enhance accuracy and stability.

8.4 Task 1: Reading Potentiometer Values via Serial Communication

In Task 1, the experiment focused on establishing one-way serial communication from the Arduino to Python by reading and transmitting potentiometer values. The Arduino converted the analog input signal into digital data using the analogRead() function and sent it to the computer via the serial port. The Python script, using the *pyserial* library, successfully received and displayed these readings in real time. As the potentiometer was adjusted, the output values on the Python console varied proportionally, confirming accurate data transmission. An LED was laterintegrated to indicate when the potentiometer reading exceeded half of its maximum value, effectively demonstrating a simple threshold control mechanism. This task provided foundational understanding of how sensor data can be captured, transmitted, and utilized for control decisions in a mechatronic system.

8.5 Task 2: Controlling Servo Motor via Serial Communication

Task 2 expanded upon the previous task by implementing bidirectional communication, allowing Python to send angle commands to the Arduino to control the servo motor. The servo motor's signal pin connected to digital pin 7, and the *Servo* library was used to position the motor according to the received data. When the user entered an angle between 0° and 180° in the Python interface, the Arduino accurately adjusted the servo position. Later, the potentiometer was integrated to enable real-time control of the servo angle, while *matplotlib* provided continuous graphical feedback of the data. Although minor response delays were noticed due to communication speed and plotting intervals, the overall system operated effectively. This task demonstrated a successful two-way interaction between hardware and software, showcasing how serial communication can be applied to real-time control and monitoring in automation and robotics systems.

8.6 Summary

The discussion confirmed that both tasks achieved their intended objectives, demonstrating the importance of serial communication in connecting embedded systems and high-level programming environments. Despite minor timing and power-related limitations, the experiment provided valuable insights into system integration, data handling, and real-time control — key competencies in mechatronics engineering.

Conclusion

The experiment successfully demonstrated the implementation of serial communication between an Arduino microcontroller and a computer using Python for real-time control of a servo motor. The main findings showed that data could be reliably transmitted from the Arduino to the computer and vice versa, allowing accurate control of the servo motor's position based on potentiometer readings or user input. The use of the *pyserial* library enabled seamless communication, while *matplotlib* provided an effective platform for real-time data visualization. The system responded smoothly and consistently to input changes, confirming the proper operation of both the hardware and software components.

The results fully supported the initial hypothesis, which proposed that establishing serial communication between the Arduino and Python would enable precise and responsive servo control. The servo motor's motion corresponded accurately to the potentiometer adjustments and Python inputs, validating the effectiveness of the design. Minor signal delays were observed due to serial data transfer speed, but these did not significantly affect performance.Overall, this experiment demonstrated the practical importance of integrating programming with embedded hardware for real-time control applications. The knowledge gained from this study has broader implications in the fields of robotics, automation, and industrial control systems, where microcontrollers are commonly used for data acquisition and actuator control. The ability to interface hardware with higher-level programming languages like Python provides a powerful platform for developing intelligent mechatronic systems, enabling innovation and adaptability in future engineering projects.

Recommendations

For future iterations of this experiment, several improvements can be implemented to enhance accuracy, functionality, and user experience. First, the hardware setup can be refined by incorporating a more stable power supply or an external power source for the servo motor. This would help prevent voltage fluctuations and ensure smoother servo movement. Additionally, using shielded jumper wires or a prototyping shield could minimize noise interference and improve the reliability of signal transmission. The software could also be improved by implementing advanced control algorithms, such as Proportional–Integral–Derivative (PID) control, to achieve more precise and stable servo positioning. Moreover, integrating a Graphical User Interface (GUI) in Python using libraries such as *Tkinter* or *PyQt* would provide an] intuitive way for users to input angles and monitor feedback.
Future versions of the experiment could also benefit from incorporating data logging features to record potentiometer values and servo angles over time for further analysis. Expanding theexperiment to include multiple servos or sensors would allow students to explore multi-axis control and sensor fusion concepts, making the setup more representative of real-world mechatronic systems. From a learning perspective, this experiment emphasized the importance of synchronization between hardware and software, accurate serial communication setup, and debugging techniques. Future students are encouraged to pay close attention to selecting the correct COM port, baud rate, and data format to ensure seamless communication. Additionally, verifying circuit connections before powering the system can prevent component damage and troubleshooting delays. Overall, careful setup, structured coding, and systematic testing were key lessons that contribute to the success of serial communication projects in mechatronics applications.

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4. *Arduino INPUT_PULLUP Explained (pinMode)*. (2021, January 11). The Robotics Back-End. https://roboticsbackend.com/arduino-input_pullup-pinmode/

Appendices

Appendix A- RAW DATA TABLES

ROTATION POSITION	ANALOG READING	CALCULATED VOLTAGE (V)
1	180	0.88
2	200	1.07
3	220	1.17

Table 1

NO	INPUT ANGLE (from python)	OBSERVED SERVO POSITION	LED INDICATOR	OBSERVATION
1	0°	Arm at lefttmost position	OFF	Servo initializes at 0°, LED off
2	45°	Arm at mid-left	OFF	Servo moves smoothly, correct position
3	90°	Arm at rightmost position	OFF	LED still off, stable mid-point
4	120°	Arm at rightmost position	ON	LED turns on after 90°
5	180°	Arm at rightmost position	ON	Maximum rotation achieved

Table 2

INPUT ANGLE(°)	EXPECTED SERVO POSITION	OBSERVED SERVO POSITION	DEVIATION(°)
0	Leftmost	Leftmost	0
45	Mid-left	Mid-left	0
90	center	center	0
120	Slightly right	Slightly right	0
150	Near rightmost	Near rightmost	0
180	rightmost	rightmost	0

Table 3 : servo motor performance

PARAMETER	EXPECTED OUTCOME	ACTUAL OBSERVATION	EVALUATION
Serial communication	continous	Stable and responsive	Achieved
Servo response	Smooth rotation	Smooth and precise	Achieved
LED logic	ON > 90°, OFF ≤ 90°	Correctly triggered	Achieved
Response time	< 0.2 s	~0.1 s	Excellent

Table 4 : system evaluation

Appendix B - CODE SNIPPET

Arduiono

```
const int potPin = A0;
const int ledPin = 7;
const float Vin = 5.0;
const float R_total = 10000;


String command = "";


void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
}
void loop()
{
  int potValue = analogRead(potPin);
  float Vout = (potValue / 1023.0) * Vin;
  float R_wiper = (Vout / Vin) * R_total;
}
```

sketch_mar13a.ino

```
#include <Servo.h>

Servo myservo;
const int potPin = A0;
const int ledPin = 7;
const float Vin = 5.0;

bool stopControl = false;
String command = "";

void setup() {
  Serial.begin(9600);
  myservo.attach(9);
  pinMode(ledPin, OUTPUT);
  digitalWrite(ledPin, LOW);
}

void loop() {
  // 1. Check for commands from Python
  if (Serial.available() > 0) {
    command = Serial.readstringUntil('\n');
    command.trim();

    if (command == "x") {
      stopControl = true;
      myservo.detach();
      digitalWrite(ledPin, LOW);
    }
    else if (command == "LED_ON" && !stopControl) {
      digitalWrite(ledPin, HIGH);
    }
    else if (command == "LED_OFF" && !stopControl) {
      digitalWrite(ledPin, LOW);
```

Python

```python
import serial
import matplotlib.pyplot as plt
import time

# Update to your actual COM port
ser = serial.Serial( port: 'COM5',  baudrate: 9600, timeout=1)
time.sleep(2)  # Wait for Arduino reset

plt.ion()
fig, ax = plt.subplots()
line, = ax.plot( *args: [], [], 'r-', label="Servo Angle")

# Setup Plot Aesthetics
ax.set_ylim( bottom: 0,  top: 180)
ax.set_xlim( left: 0,  right: 100)
ax.set_title("Real-Time Mechatronics Data")
ax.set_xlabel("Samples")
ax.set_ylabel("Angle (°)")
ax.legend()

x_vals, y_vals = [], []

print("Running... Data parsing & Plotting active. Press CTRL+C to stop.")

try:
    while True:
        line_raw = ser.readline().decode().strip()

        # Check if line has the "/" format from Code 3
        if "/" in line_raw:
            try:
```


```python
import serial
import matplotlib.pyplot as plt
import time

# Update to your actual COM port
ser = serial.Serial( port: 'COM5',  baudrate: 9600, timeout=1)
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x_vals, y_vals = [], []

print("Running... Data parsing & Plotting active. Press CTRL+C to stop.")

try:
    while True:
        line_raw = ser.readline().decode().strip()

        # Check if line has the "/" format from Code 3
        if "/" in line_raw:
            try:
```

```python
            try:
                # 1. Parse Data
                parts = line_raw.split("/")
                voltage = float(parts[0])
                adc_val = int(parts[1].replace("ADC:", "").strip())
                angle = int(parts[3]) if len(parts) > 3 else int(map(adc_val, 0, 1023, 0, 180))  # Backup calc

                print(f"V: {voltage:.2f}V | ADC: {adc_val} | Angle: {angle}°")

                # 2. Update Plot Data
                x_vals.append(len(x_vals))
                y_vals.append(angle)
                line.set_data(x_vals, y_vals)

                if len(x_vals) > 100:
                    ax.set_xlim(len(x_vals) - 100, len(x_vals))

                # 3. LED Control Logic (Bidirectional)
                if adc_val > 512:
                    ser.write(b"LED_ON\n")
                else:
                    ser.write(b"LED_OFF\n")

                plt.pause(0.01)

            except Exception as e:
                print(f"Parsing Error: {e}")
```

```python
                if len(x_vals) > 100:
                    ax.set_xlim(len(x_vals) - 100, len(x_vals))

                # 3. LED Control Logic (Bidirectional)
                if adc_val > 512:
                    ser.write(b"LED_ON\n")
                else:
                    ser.write(b"LED_OFF\n")

                plt.pause(0.01)

            except Exception as e:
                print(f"Parsing Error: {e}")

    except KeyboardInterrupt:
        print("\nStopping... Sending shutdown command.")
        ser.write(b'x')
        time.sleep(0.5)
    finally:
        ser.close()
        plt.ioff()
        plt.show()
        print("Serial connection closed.")
```

Acknowledgments

We would like to express my sincere gratitude to my lecturer and lab instructor for their continuous guidance, clear explanations, and assistance with experimental setups throughout this experiment. Their detailed instructions with the several needed files attached have helped us understand how to properly set up the Cytron CT-UNO microcontroller, connect the 7-segment display, and program the system using Arduino IDE.

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Overall, the success of this experiment would not have been possible without the guidance and support from both the instructors and my lab partners.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or a person.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

Signature:

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Understand /

Agree /

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